

Trapezoidal Rule of Integration

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What is Integration

Integration:

The process of measuring the area under a function plotted on a graph.

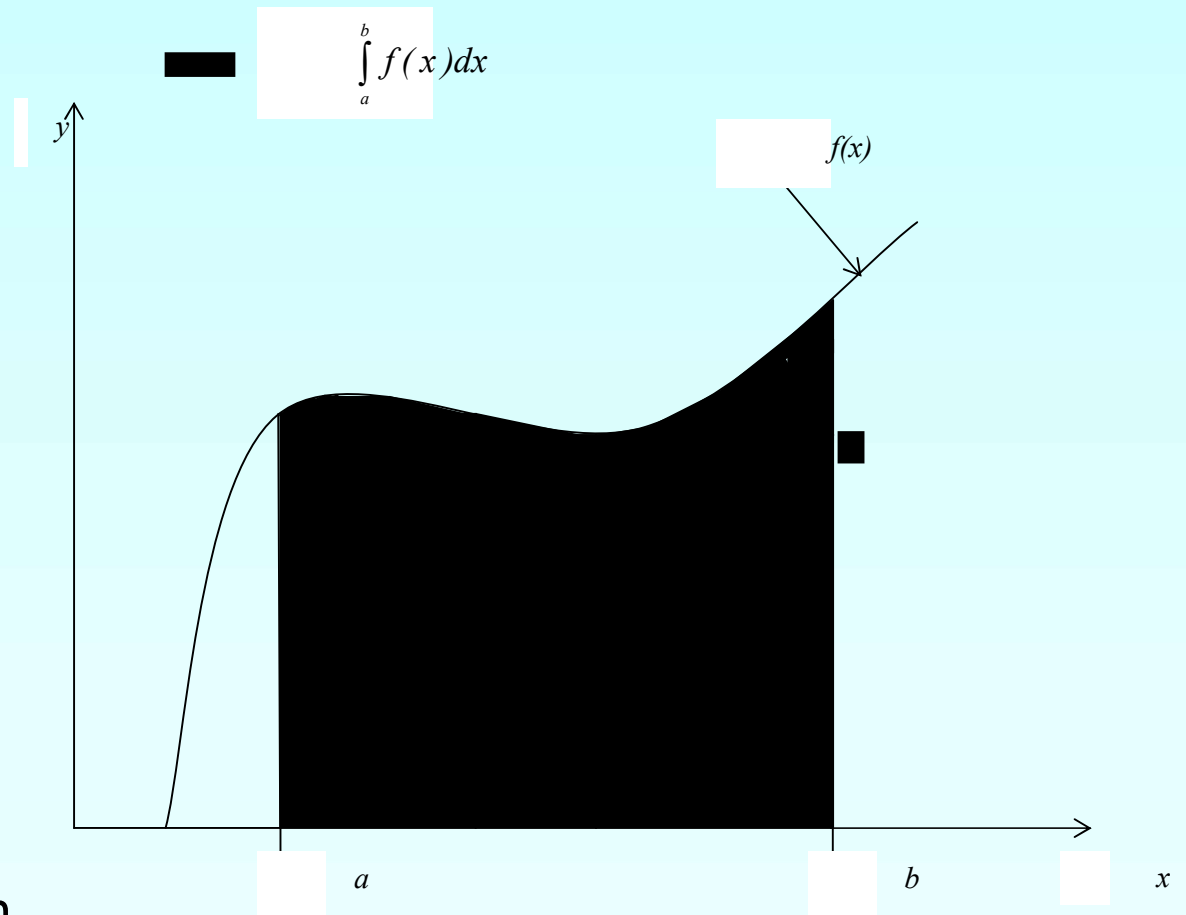
$$I = \int_a^b f(x) dx$$

Where:

$f(x)$ is the integrand

a = lower limit of integration

b = upper limit of integration



Basis of Trapezoidal Rule

Trapezoidal Rule is based on the Newton-Cotes Formula that states if one can approximate the integrand as an n^{th} order polynomial...

$$I = \int_a^b f(x) dx \quad \text{where} \quad f(x) \approx f_n(x)$$

$$\text{and} \quad f_n(x) = a_0 + a_1x + \dots + a_{n-1}x^{n-1} + a_nx^n$$

Basis of Trapezoidal Rule

Then the integral of that function is approximated by the integral of that n^{th} order polynomial.

$$\int_a^b f(x) \approx \int_a^b f_n(x)$$

Trapezoidal Rule assumes $n=1$, that is, the area under the linear polynomial,

$$\int_a^b f(x)dx = (b-a) \left[\frac{f(a) + f(b)}{2} \right]$$

Derivation of the Trapezoidal Rule

Method Derived From Geometry

The area under the curve is a trapezoid.
The integral

$$\begin{aligned} \int_a^b f(x) dx &\approx \text{Area of trapezoid} \\ &= \frac{1}{2} (\text{Sum of parallel sides})(\text{height}) \\ &= \frac{1}{2} (f(b) + f(a))(b - a) \\ &= (b - a) \left[\frac{f(a) + f(b)}{2} \right] \end{aligned}$$

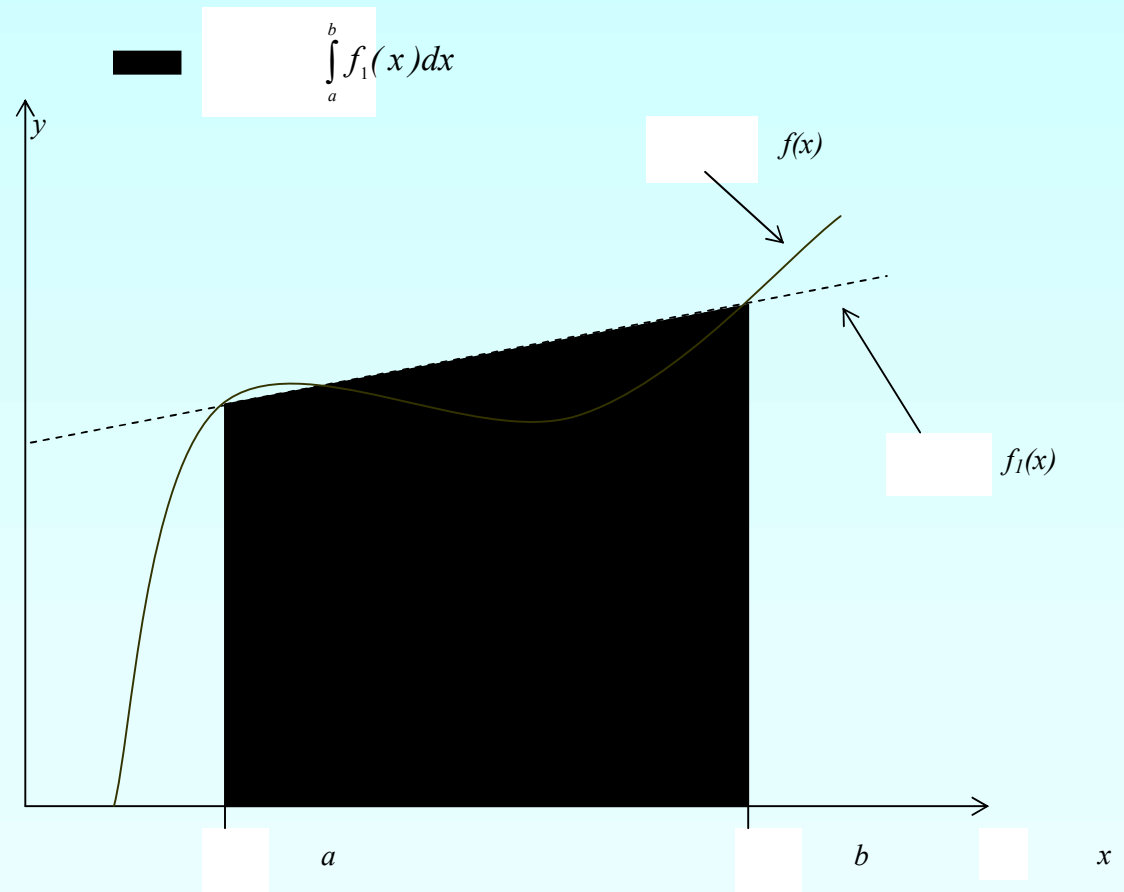


Figure 2: Geometric Representation

Example 1

The vertical distance covered by a rocket from $t=8$ to $t=30$ seconds is given by:

$$x = \int_8^{30} \left(2000 \ln \left[\frac{140000}{140000 - 2100t} \right] - 9.8t \right) dt$$

- a) Use single segment Trapezoidal rule to find the distance covered.
- b) Find the true error, E_t for part (a).
- c) Find the absolute relative true error, $|\epsilon_a|$ for part (a).

Solution

$$\text{a)} \quad I \approx (b - a) \left[\frac{f(a) + f(b)}{2} \right]$$

$$a = 8 \quad b = 30$$

$$f(t) = 2000 \ln \left[\frac{140000}{140000 - 2100t} \right] - 9.8t$$

$$f(8) = 2000 \ln \left[\frac{140000}{140000 - 2100(8)} \right] - 9.8(8) = 177.27 \text{ m/s}$$

$$f(30) = 2000 \ln \left[\frac{140000}{140000 - 2100(30)} \right] - 9.8(30) = 901.67 \text{ m/s}$$

Solution (cont)

$$\begin{aligned} \text{a)} \quad I &= (30 - 8) \left[\frac{177.27 + 901.67}{2} \right] \\ &= 11868 \text{ m} \end{aligned}$$

b) The exact value of the above integral is

$$x = \int_8^{30} \left(2000 \ln \left[\frac{140000}{140000 - 2100t} \right] - 9.8t \right) dt = 11061 \text{ m}$$

Solution (cont)

b)

$$\begin{aligned} E_t &= \text{True Value} - \text{Approximate Value} \\ &= 11061 - 11868 \\ &= -807 \text{ m} \end{aligned}$$

c) The absolute relative true error, $|\epsilon_t|$, would be

$$|\epsilon_t| = \left| \frac{11061 - 11868}{11061} \right| \times 100 = 7.2959\%$$

Multiple Segment Trapezoidal Rule

In Example 1, the true error using single segment trapezoidal rule was large. We can divide the interval $[8,30]$ into $[8,19]$ and $[19,30]$ intervals and apply Trapezoidal rule over each segment.

$$f(t) = 2000 \ln \left(\frac{140000}{140000 - 2100t} \right) - 9.8t$$

$$\int_8^{30} f(t) dt = \int_8^{19} f(t) dt + \int_{19}^{30} f(t) dt$$

$$= (19 - 8) \left[\frac{f(8) + f(19)}{2} \right] + (30 - 19) \left[\frac{f(19) + f(30)}{2} \right]$$

Multiple Segment Trapezoidal Rule

With

$$f(8) = 177.27 \text{ m/s}$$

$$f(30) = 901.67 \text{ m/s}$$

$$f(19) = 484.75 \text{ m/s}$$

Hence:

$$\int_8^{30} f(t) dt = (19 - 8) \left[\frac{177.27 + 484.75}{2} \right] + (30 - 19) \left[\frac{484.75 + 901.67}{2} \right]$$

$$= 11266 \text{ m}$$

Multiple Segment Trapezoidal Rule

The true error is:

$$\begin{aligned} E_t &= 11061 - 11266 \\ &= -205 \text{ m} \end{aligned}$$

The true error now is reduced from -807 m to -205 m.

Extending this procedure to divide the interval into equal segments to apply the Trapezoidal rule; the sum of the results obtained for each segment is the approximate value of the integral.

Multiple Segment Trapezoidal Rule

Divide into equal segments as shown in Figure 4. Then the width of each segment is:

$$h = \frac{b - a}{n}$$

The integral I is:

$$I = \int_a^b f(x) dx$$

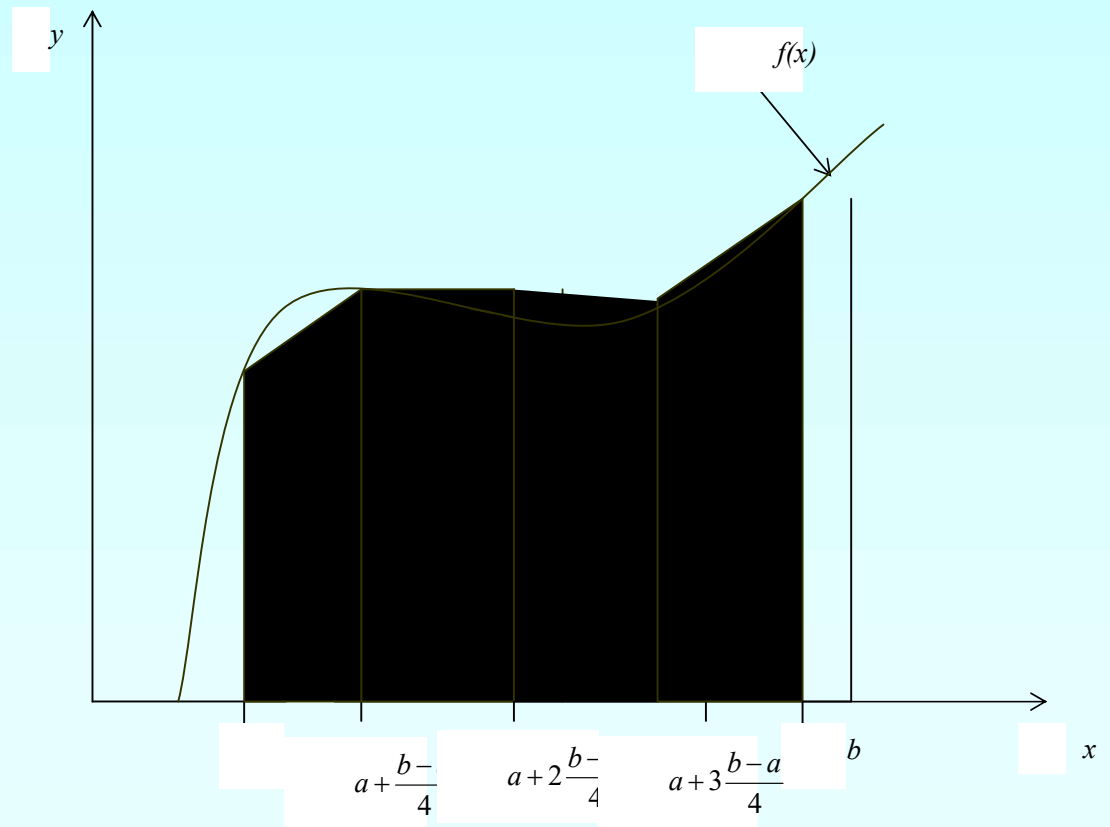


Figure 4: Multiple ($n=4$) Segment Trapezoidal Rule

Multiple Segment Trapezoidal Rule

The integral I can be broken into n integrals as:

$$\int_a^b f(x)dx = \int_a^{a+h} f(x)dx + \int_{a+h}^{a+2h} f(x)dx + \dots + \int_{a+(n-2)h}^{a+(n-1)h} f(x)dx + \int_{a+(n-1)h}^b f(x)dx$$

Applying Trapezoidal rule on each segment gives:

$$\int_a^b f(x)dx = \frac{b-a}{2n} \left[f(a) + 2 \left\{ \sum_{i=1}^{n-1} f(a+ih) \right\} + f(b) \right]$$

Example 2

The vertical distance covered by a rocket from $t = 8$ to $t = 30$ seconds is given by:

$$x = \int_8^{30} \left(2000 \ln \left[\frac{140000}{140000 - 2100t} \right] - 9.8t \right) dt$$

- a) Use two-segment Trapezoidal rule to find the distance covered.
- b) Find the true error, E_t for part (a).
- c) Find the absolute relative true error, $|\epsilon_a|$ for part (a).

Solution

a) The solution using 2-segment Trapezoidal rule is

$$I = \frac{b-a}{2n} \left[f(a) + 2 \left\{ \sum_{i=1}^{n-1} f(a+ih) \right\} + f(b) \right]$$

$$n = 2 \qquad a = 8 \qquad b = 30$$

$$h = \frac{b-a}{n} = \frac{30-8}{2} = 11$$

Solution (cont)

Then:

$$\begin{aligned} I &= \frac{30 - 8}{2(2)} \left[f(8) + 2 \left\{ \sum_{i=1}^{2-1} f(a + ih) \right\} + f(30) \right] \\ &= \frac{22}{4} [f(8) + 2f(19) + f(30)] \\ &= \frac{22}{4} [177.27 + 2(484.75) + 901.67] \\ &= 11266 \text{ m} \end{aligned}$$

Solution (cont)

b) The exact value of the above integral is

$$x = \int_8^{30} \left(2000 \ln \left[\frac{140000}{140000 - 2100t} \right] - 9.8t \right) dt = 11061 \text{ m}$$

so the true error is

$$\begin{aligned} E_t &= \text{True Value} - \text{Approximate Value} \\ &= 11061 - 11266 \end{aligned}$$

Solution (cont)

The absolute relative true error, $|\epsilon_t|$, would be

$$|\epsilon_t| = \left| \frac{\text{True Error}}{\text{True Value}} \right| \times 100$$

$$= \left| \frac{11061 - 11266}{11061} \right| \times 100$$

$$= 1.8534\%$$

Solution (cont)

Table 1 gives the values obtained using multiple segment Trapezoidal rule for:

$$x = \int_8^{30} \left(2000 \ln \left[\frac{140000}{140000 - 2100t} \right] - 9.8t \right) dt$$

n	Value	E_t	 ϵ_t %	 ϵ_a %
1	11868	-807	7.296	---
2	11266	-205	1.853	5.343
3	11153	-91.4	0.8265	1.019
4	11113	-51.5	0.4655	0.3594
5	11094	-33.0	0.2981	0.1669
6	11084	-22.9	0.2070	0.09082
7	11078	-16.8	0.1521	0.05482
8	11074	-12.9	0.1165	0.03560

Table 1: Multiple Segment Trapezoidal Rule Values

Example 3

Use Multiple Segment Trapezoidal Rule to find the area under the curve

$$f(x) = \frac{300x}{1 + e^x} \quad \text{from } x = 0 \quad \text{to} \quad x = 10$$

Using two segments, we get $h = \frac{10 - 0}{2} = 5$ and

$$f(0) = \frac{300(0)}{1 + e^0} = 0 \quad f(5) = \frac{300(5)}{1 + e^5} = 10.039 \quad f(10) = \frac{300(10)}{1 + e^{10}} = 0.136$$

Solution

Then:

$$\begin{aligned} I &= \frac{b-a}{2n} \left[f(a) + 2 \left\{ \sum_{i=1}^{n-1} f(a+ih) \right\} + f(b) \right] \\ &= \frac{10-0}{2(2)} \left[f(0) + 2 \left\{ \sum_{i=1}^{2-1} f(0+5) \right\} + f(10) \right] \\ &= \frac{10}{4} [f(0) + 2f(5) + f(10)] = \frac{10}{4} [0 + 2(10.039) + 0.136] \\ &= 50.535 \end{aligned}$$

Solution (cont)

So what is the true value of this integral?

$$\int_0^{10} \frac{300x}{1+e^x} dx = 246.59$$

Making the absolute relative true error:

$$\begin{aligned} |\epsilon_t| &= \left| \frac{246.59 - 50.535}{246.59} \right| \times 100\% \\ &= 79.506\% \end{aligned}$$

Solution (cont)

Table 2: Values obtained using Multiple Segment Trapezoidal Rule for:

$$\int_0^{10} \frac{300x}{1+e^x} dx$$

n	Approximate Value	E_t	$ \epsilon_t $
1	0.681	245.91	99.724%
2	50.535	196.05	79.505%
4	170.61	75.978	30.812%
8	227.04	19.546	7.927%
16	241.70	4.887	1.982%
32	245.37	1.222	0.495%
64	246.28	0.305	0.124%

Error in Multiple Segment Trapezoidal Rule

The true error for a single segment Trapezoidal rule is given by:

$$E_t = \frac{(b-a)^3}{12} f''(\zeta), \quad a < \zeta < b \quad \text{where } \zeta \text{ is some point in } [a, b]$$

What is the error, then in the multiple segment Trapezoidal rule? It will be simply the sum of the errors from each segment, where the error in each segment is that of the single segment Trapezoidal rule.

The error in each segment is

$$\begin{aligned} E_1 &= \frac{[(a+h)-a]^3}{12} f''(\zeta_1), \quad a < \zeta_1 < a+h \\ &= \frac{h^3}{12} f''(\zeta_1) \end{aligned}$$

Error in Multiple Segment Trapezoidal Rule

Similarly:

$$E_i = \frac{[(a + ih) - (a + (i - 1)h)]^3}{12} f''(\zeta_i), \quad a + (i - 1)h < \zeta_i < a + ih$$
$$= \frac{h^3}{12} f''(\zeta_i)$$

It then follows that:

$$E_n = \frac{[b - \{a + (n - 1)h\}]^3}{12} f''(\zeta_n), \quad a + (n - 1)h < \zeta_n < b$$
$$= \frac{h^3}{12} f''(\zeta_n)$$

Error in Multiple Segment Trapezoidal Rule

Hence the total error in multiple segment Trapezoidal rule is

$$E_t = \sum_{i=1}^n E_i = \frac{h^3}{12} \sum_{i=1}^n f''(\zeta_i) = \frac{(b-a)^3}{12n^2} \frac{\sum_{i=1}^n f''(\zeta_i)}{n}$$

The term $\frac{\sum_{i=1}^n f''(\zeta_i)}{n}$ is an approximate average value of the $f''(x)$, $a < x < b$

Hence:

$$E_t = \frac{(b-a)^3}{12n^2} \frac{\sum_{i=1}^n f''(\zeta_i)}{n}$$

Error in Multiple Segment Trapezoidal Rule

Below is the table for the integral $\int_8^{30} \left(2000 \ln \left[\frac{140000}{140000 - 2100t} \right] - 9.8t \right) dt$

as a function of the number of segments. You can visualize that as the number of segments are doubled, the true error gets approximately quartered.

n	Value	E_t	$ \epsilon_t \%$	$ \epsilon_a \%$
2	11266	-205	1.854	5.343
4	11113	-51.5	0.4655	0.3594
8	11074	-12.9	0.1165	0.03560
16	11065	-3.22	0.02913	0.00401

Simpson's $1/3^{\text{rd}}$ Rule of Integration

What is Integration?

Integration

The process of measuring the area under a curve.

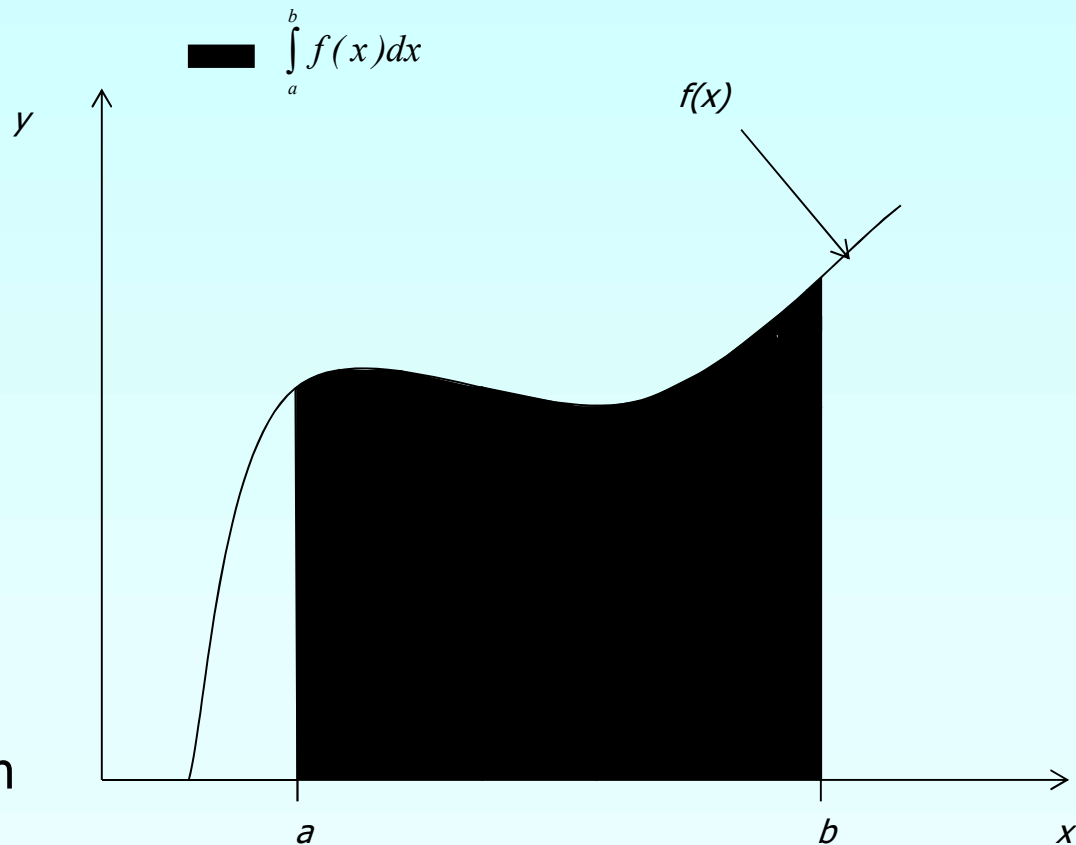
$$I = \int_a^b f(x) dx$$

Where:

$f(x)$ is the integrand

a = lower limit of integration

b = upper limit of integration



Simpson's $1/3^{\text{rd}}$ Rule

Basis of Simpson's 1/3rd Rule

Trapezoidal rule was based on approximating the integrand by a first order polynomial, and then integrating the polynomial in the interval of integration. Simpson's 1/3rd rule is an extension of Trapezoidal rule where the integrand is approximated by a second order polynomial.

Hence

$$I = \int_a^b f(x)dx \approx \int_a^b f_2(x)dx$$

Where $f_2(x)$ is a second order polynomial.

$$f_2(x) = a_0 + a_1x + a_2x^2$$

Basis of Simpson's 1/3rd Rule

Choose

$$(a, f(a)), \left(\frac{a+b}{2}, f\left(\frac{a+b}{2}\right) \right), \text{ and } (b, f(b))$$

as the three points of the function to evaluate a_0 , a_1 and a_2 .

$$f(a) = f_2(a) = a_0 + a_1a + a_2a^2$$

$$f\left(\frac{a+b}{2}\right) = f_2\left(\frac{a+b}{2}\right) = a_0 + a_1\left(\frac{a+b}{2}\right) + a_2\left(\frac{a+b}{2}\right)^2$$

$$f(b) = f_2(b) = a_0 + a_1b + a_2b^2$$

Basis of Simpson's 1/3rd Rule

Solving the previous equations for a_0 , a_1 and a_2 give

$$a_0 = \frac{a^2 f(b) + abf(b) - 4abf\left(\frac{a+b}{2}\right) + abf(a) + b^2 f(a)}{a^2 - 2ab + b^2}$$

$$a_1 = -\frac{af(a) - 4af\left(\frac{a+b}{2}\right) + 3af(b) + 3bf(a) - 4bf\left(\frac{a+b}{2}\right) + bf(b)}{a^2 - 2ab + b^2}$$

$$a_2 = \frac{2\left(f(a) - 2f\left(\frac{a+b}{2}\right) + f(b)\right)}{a^2 - 2ab + b^2}$$

Basis of Simpson's 1/3rd Rule

Then

$$\begin{aligned} I &\approx \int_a^b f_2(x) dx \\ &= \int_a^b (a_0 + a_1 x + a_2 x^2) dx \\ &= \left[a_0 x + a_1 \frac{x^2}{2} + a_2 \frac{x^3}{3} \right]_a^b \\ &= a_0(b-a) + a_1 \frac{b^2 - a^2}{2} + a_2 \frac{b^3 - a^3}{3} \end{aligned}$$

Basis of Simpson's 1/3rd Rule

Substituting values of a_0 , a_1 , a_2 give

$$\int_a^b f_2(x) dx = \frac{b-a}{6} \left[f(a) + 4f\left(\frac{a+b}{2}\right) + f(b) \right]$$

Since for Simpson's 1/3rd Rule, the interval $[a, b]$ is broken into 2 segments, the segment width

$$h = \frac{b-a}{2}$$

Basis of Simpson's 1/3rd Rule

Hence

$$\int_a^b f_2(x) dx = \frac{h}{3} \left[f(a) + 4f\left(\frac{a+b}{2}\right) + f(b) \right]$$

Because the above form has 1/3 in its formula, it is called Simpson's 1/3rd Rule.

Example 1

The distance covered by a rocket from $t=8$ to $t=30$ is given by

$$x = \int_8^{30} \left(2000 \ln \left[\frac{140000}{140000 - 2100t} \right] - 9.8t \right) dt$$

- a) Use Simpson's 1/3rd Rule to find the approximate value of x
- b) Find the true error, E_t
- c) Find the absolute relative true error, $|\epsilon_t|$

Solution

a)

$$\begin{aligned}x &= \int_8^{30} f(t) dt \\x &= \left(\frac{b-a}{6}\right) \left[f(a) + 4f\left(\frac{a+b}{2}\right) + f(b) \right] \\&= \left(\frac{30-8}{6}\right) [f(8) + 4f(19) + f(30)] \\&= \left(\frac{22}{6}\right) [177.2667 + 4(484.7455) + 901.6740] \\&= 11065.72 \text{ m}\end{aligned}$$

Solution (cont)

b) The exact value of the above integral is

$$x = \int_8^{30} \left(2000 \ln \left[\frac{140000}{140000 - 2100t} \right] - 9.8t \right) dt$$

$$= 11061.34 \text{ m}$$

True Error

$$E_t = 11061.34 - 11065.72$$

$$= -4.38 \text{ m}$$

Solution (cont)

a)c) Absolute relative true error,

$$|\epsilon_t| = \left| \frac{11061.34 - 11065.72}{11061.34} \right| \times 100\%$$
$$= 0.0396\%$$

Multiple Segment Simpson's 1/3rd Rule

Multiple Segment Simpson's 1/3rd Rule

Just like in multiple segment Trapezoidal Rule, one can subdivide the interval $[a, b]$ into n segments and apply Simpson's 1/3rd Rule repeatedly over every two segments. Note that n needs to be even. Divide interval $[a, b]$ into equal segments, hence the segment width

$$h = \frac{b - a}{n} \qquad \int_a^b f(x) dx = \int_{x_0}^{x_n} f(x) dx$$

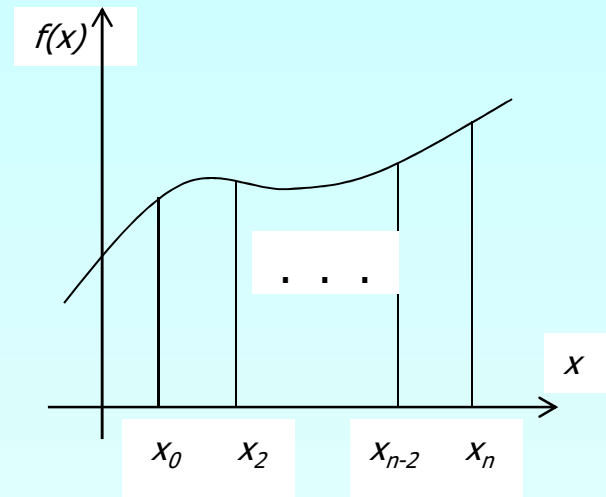
where

$$x_0 = a \qquad x_n = b$$

Multiple Segment Simpson's 1/3rd Rule

$$\int_a^b f(x) dx = \int_{x_0}^{x_2} f(x) dx + \int_{x_2}^{x_4} f(x) dx + \dots$$

$$\dots + \int_{x_{n-4}}^{x_{n-2}} f(x) dx + \int_{x_{n-2}}^{x_n} f(x) dx$$



Apply Simpson's 1/3rd Rule over each interval,

$$\int_a^b f(x) dx = (x_2 - x_0) \left[\frac{f(x_0) + 4f(x_1) + f(x_2)}{6} \right] + \dots$$

$$+ (x_4 - x_2) \left[\frac{f(x_2) + 4f(x_3) + f(x_4)}{6} \right] + \dots$$

Multiple Segment Simpson's 1/3rd Rule

$$\begin{aligned} & \dots + (x_{n-2} - x_{n-4}) \left[\frac{f(x_{n-4}) + 4f(x_{n-3}) + f(x_{n-2})}{6} \right] + \dots \\ & + (x_n - x_{n-2}) \left[\frac{f(x_{n-2}) + 4f(x_{n-1}) + f(x_n)}{6} \right] \end{aligned}$$

Since

$$x_i - x_{i-2} = 2h \qquad i = 2, 4, \dots, n$$

Multiple Segment Simpson's 1/3rd Rule

Then

$$\begin{aligned}\int_a^b f(x)dx &= 2h \left[\frac{f(x_0) + 4f(x_1) + f(x_2)}{6} \right] + \dots \\ &+ 2h \left[\frac{f(x_2) + 4f(x_3) + f(x_4)}{6} \right] + \dots \\ &+ 2h \left[\frac{f(x_{n-4}) + 4f(x_{n-3}) + f(x_{n-2})}{6} \right] + \dots \\ &+ 2h \left[\frac{f(x_{n-2}) + 4f(x_{n-1}) + f(x_n)}{6} \right]\end{aligned}$$

Multiple Segment Simpson's 1/3rd Rule

$$\begin{aligned}\int_a^b f(x)dx &= \frac{h}{3} [f(x_0) + 4\{f(x_1) + f(x_3) + \dots + f(x_{n-1})\} + \dots] \\ &\quad \dots + 2\{f(x_2) + f(x_4) + \dots + f(x_{n-2})\} + f(x_n)] \\ &= \frac{h}{3} \left[f(x_0) + 4 \sum_{\substack{i=1 \\ i=\text{odd}}}^{n-1} f(x_i) + 2 \sum_{\substack{i=2 \\ i=\text{even}}}^{n-2} f(x_i) + f(x_n) \right] \\ &= \frac{b-a}{3n} \left[f(x_0) + 4 \sum_{\substack{i=1 \\ i=\text{odd}}}^{n-1} f(x_i) + 2 \sum_{\substack{i=2 \\ i=\text{even}}}^{n-2} f(x_i) + f(x_n) \right]\end{aligned}$$

Example 2

Use 4-segment Simpson's 1/3rd Rule to approximate the distance covered by a rocket from $t=8$ to $t=30$ as given by

$$x = \int_8^{30} \left(2000 \ln \left[\frac{140000}{140000 - 2100t} \right] - 9.8t \right) dt$$

- a) Use four segment Simpson's 1/3rd Rule to find the approximate value of x .
- b) Find the true error, E_t for part (a).
- c) Find the absolute relative true error, $|\epsilon_a|$ for part (a).

Solution

a) Using n segment Simpson's 1/3rd Rule,

$$h = \frac{30 - 8}{4} = 5.5$$

So

$$f(t_0) = f(8)$$

$$f(t_1) = f(8 + 5.5) = f(13.5)$$

$$f(t_2) = f(13.5 + 5.5) = f(19)$$

$$f(t_3) = f(19 + 5.5) = f(24.5)$$

$$f(t_4) = f(30)$$

Solution (cont.)

$$\begin{aligned}x &= \frac{b-a}{3n} \left[f(t_0) + 4 \sum_{\substack{i=1 \\ i=\text{odd}}}^{n-1} f(t_i) + 2 \sum_{\substack{i=2 \\ i=\text{even}}}^{n-2} f(t_i) + f(t_n) \right] \\&= \frac{30-8}{3(4)} \left[f(8) + 4 \sum_{\substack{i=1 \\ i=\text{odd}}}^3 f(t_i) + 2 \sum_{\substack{i=2 \\ i=\text{even}}}^2 f(t_i) + f(30) \right] \\&= \frac{22}{12} [f(8) + 4f(t_1) + 4f(t_3) + 2f(t_2) + f(30)]\end{aligned}$$

Solution (cont.)

cont.

$$= \frac{11}{6} [f(8) + 4f(13.5) + 4f(24.5) + 2f(19) + f(30)]$$

$$= \frac{11}{6} [177.2667 + 4(320.2469) + 4(676.0501) + 2(484.7455) + 901.6740]$$

$$= 11061.64 \text{ m}$$

Solution (cont.)

- b) In this case, the true error is

$$E_t = 11061.34 - 11061.64 = -0.30 \text{ m}$$

- c) The absolute relative true error

$$\begin{aligned} |\epsilon_t| &= \left| \frac{11061.34 - 11061.64}{11061.34} \right| \times 100\% \\ &= 0.0027\% \end{aligned}$$

Solution (cont.)

Table 1: Values of Simpson's 1/3rd Rule for Example 2 with multiple segments

n	Approximate Value	E_t	$ \epsilon_t $
2	11065.72	4.38	0.0396%
4	11061.64	0.30	0.0027%
6	11061.40	0.06	0.0005%
8	11061.35	0.01	0.0001%
10	11061.34	0.00	0.0000%

Error in the Multiple Segment Simpson's 1/3rd Rule

The true error in a single application of Simpson's 1/3rd Rule is given as

$$E_t = -\frac{(b-a)^5}{2880} f^{(4)}(\zeta), \quad a < \zeta < b$$

In Multiple Segment Simpson's 1/3rd Rule, the error is the sum of the errors in each application of Simpson's 1/3rd Rule. The error in n segment Simpson's 1/3rd Rule is given by

$$E_1 = -\frac{(x_2 - x_0)^5}{2880} f^{(4)}(\zeta_1) = -\frac{h^5}{90} f^{(4)}(\zeta_1), \quad x_0 < \zeta_1 < x_2$$
$$E_2 = -\frac{(x_4 - x_2)^5}{2880} f^{(4)}(\zeta_2) = -\frac{h^5}{90} f^{(4)}(\zeta_2), \quad x_2 < \zeta_2 < x_4$$

Error in the Multiple Segment Simpson's 1/3rd Rule

$$E_i = -\frac{(x_{2i} - x_{2(i-1)})^5}{2880} f^{(4)}(\zeta_i) = -\frac{h^5}{90} f^{(4)}(\zeta_i), \quad x_{2(i-1)} < \zeta_i < x_{2i}$$

⋮

$$E_{\frac{n}{2}-1} = -\frac{(x_{n-2} - x_{n-4})^5}{2880} f^{(4)}\left(\zeta_{\frac{n}{2}-1}\right) = -\frac{h^5}{90} f^{(4)}\left(\zeta_{\frac{n}{2}-1}\right), \quad x_{n-4} < \zeta_{\frac{n}{2}-1} < x_{n-2}$$

$$E_{\frac{n}{2}} = -\frac{(x_n - x_{n-2})^5}{2880} f^{(4)}\left(\zeta_{\frac{n}{2}}\right) = -\frac{h^5}{90} f^{(4)}\left(\zeta_{\frac{n}{2}}\right), \quad x_{n-2} < \zeta_{\frac{n}{2}} < x_n$$

Error in the Multiple Segment Simpson's 1/3rd Rule

Hence, the total error in Multiple Segment Simpson's 1/3rd Rule is

$$\begin{aligned} E_t = \sum_{i=1}^{\frac{n}{2}} E_i &= -\frac{h^5}{90} \sum_{i=1}^{\frac{n}{2}} f^{(4)}(\zeta_i) = -\frac{(b-a)^5}{90n^5} \sum_{i=1}^{\frac{n}{2}} f^{(4)}(\zeta_i) \\ &= -\frac{(b-a)^5}{90n^4} \frac{\sum_{i=1}^{\frac{n}{2}} f^{(4)}(\zeta_i)}{n} \end{aligned}$$

Error in the Multiple Segment Simpson's 1/3rd Rule

The term $\frac{\sum_{i=1}^{\frac{n}{2}} f^{(4)}(\zeta_i)}{n}$ is an approximate average value of

$$f^{(4)}(x), a < x < b$$

Hence

$$E_t = -\frac{(b-a)^5}{90n^4} \bar{f}^{(4)}$$

where

$$\bar{f}^{(4)} = \frac{\sum_{i=1}^{\frac{n}{2}} f^{(4)}(\zeta_i)}{n}$$